



November 18–19 2015, Santa Clara

WELCOME

Raghu Das, CEO, IDTechEx



Part of the IDTechEx
Global Event Series

Next IDTechEx Show!:
27-28 April 2016 / Berlin, Germany

Co-located events bring together overlapping emerging technologies and their supply chains

New Materials, Manufacturing and Components

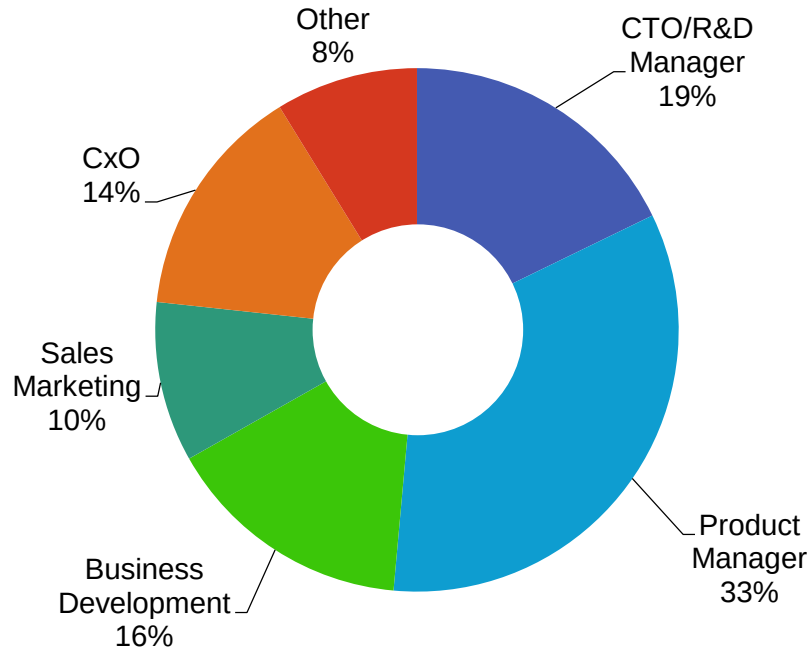


Emerging Platforms

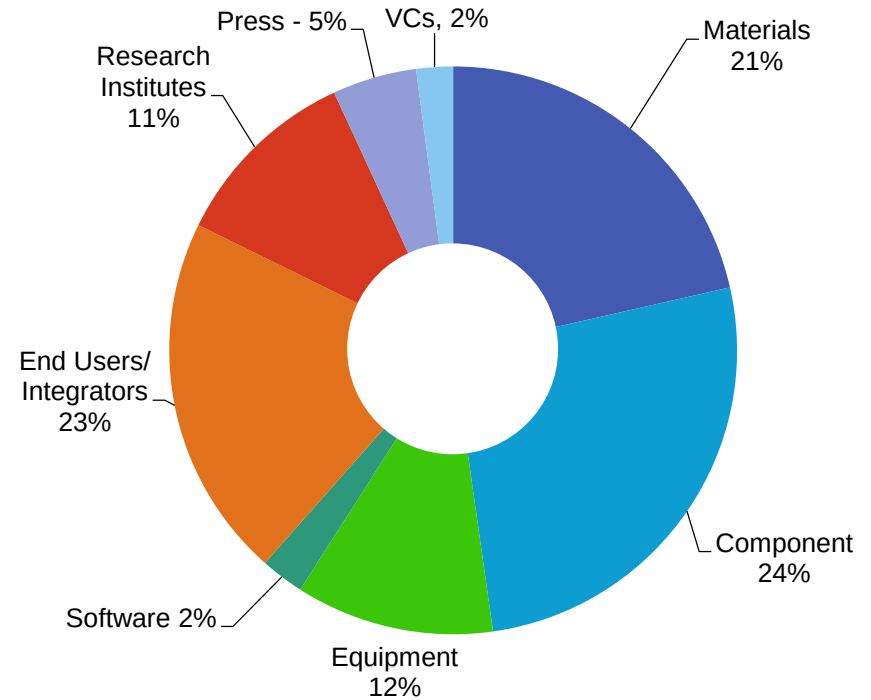


Event Statistics

Attendees by job function



Attendees by value chain positioning



- Attendees from **42 countries**
- **18% more** attendees pre-registered than 2014
- **218 exhibitors**

Lead Sponsor



NovaCentrix is focused on technologies and materials to enable manufacturing in Printed Electronics.

The PulseForge® photonic curing tools process high-temperature materials on low-temperature substrates.

Booth C06

Gold Sponsors



Booth A18



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Booth G22

Session Sponsors



Booth V24



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Booth G06



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Booth O33



Booth R31



Booth N21



Booth C04



Booth W28



Booth M27

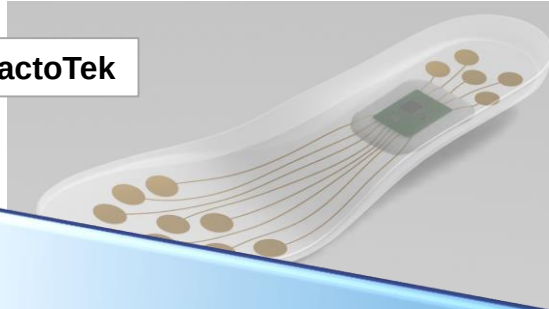


Booth T33

Ten Technology Highlights at the IDTechEx Show!

1. Structural Electronics

TactoTek



T-ink



Volvo



1915
Components-in-a-box

2015
Structural electronics / smart materials

2115



Voxel8



VTT

2. Many New Printed Electronics Products Arrive

Supplier push

Chemical, ink, film
and paper
suppliers **27%**

Production
machinery and
know how **12%**

Many available
components **58%:**
Displays
Sensors
Batteries
Photovoltaic cells
Transistors
Memory
Actuators

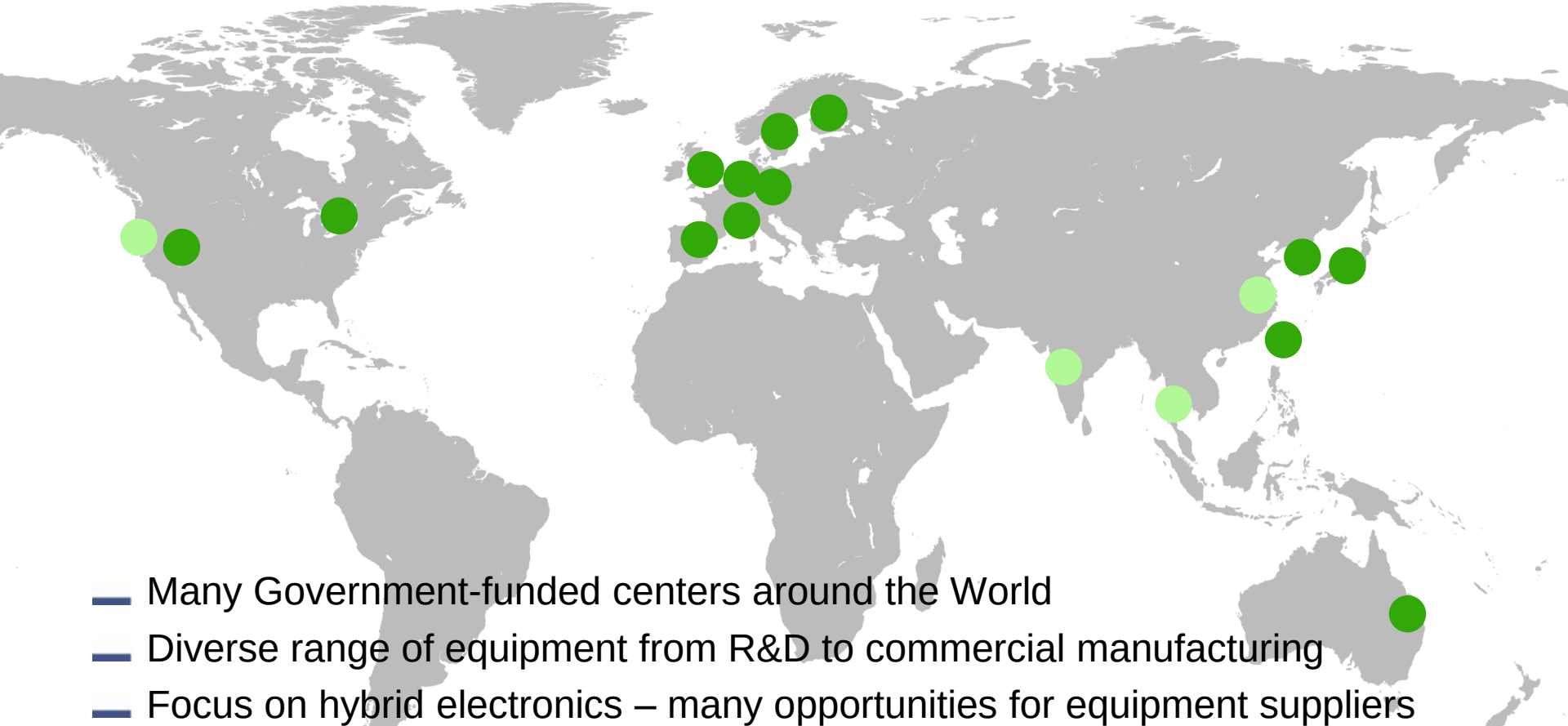
Integration and
product
manufacture
3%

User pull

End Markets

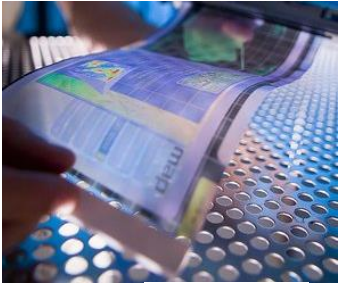
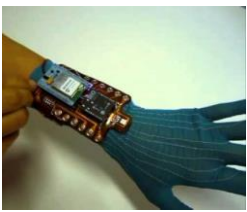


3. Printed & Flexible Electronics Manufacturing Comes Center Stage



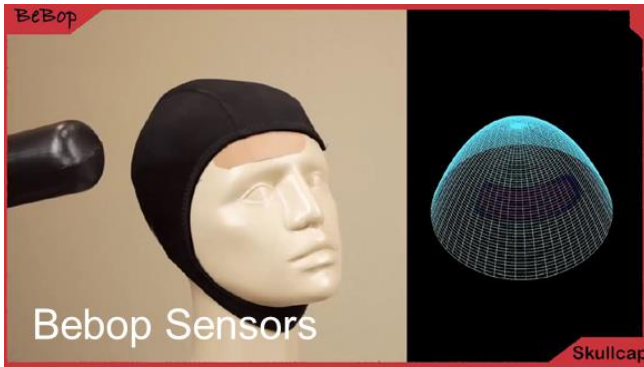
- Many Government-funded centers around the World
- Diverse range of equipment from R&D to commercial manufacturing
- Focus on hybrid electronics – many opportunities for equipment suppliers
- Printing in production: barrier layers for OLEDs, PV bus bars...
- Large EMS companies now very active with flexible/printed electronics

4. Wearable Tech Becomes Wearable

Trend	Examples
Miniaturisation, and form factors moving from bulky/rigid to thin/ flexible	  
Ugly technology becoming more fashionable and/or invisible	  
Increasing prioritisation of user comfort and convenience	
Improving user interfaces with computers	
Increasingly health , environment and safety conscious society	

- Health and fitness will see strongest growth, infotainment will be largest.
- Some prone to rapid commoditization due to well-established supply chains. Developing strong use cases, regulatory intervention, differentiation with form factors and new technologies and branding will be the countermeasure.

5. Stretchable Electronics



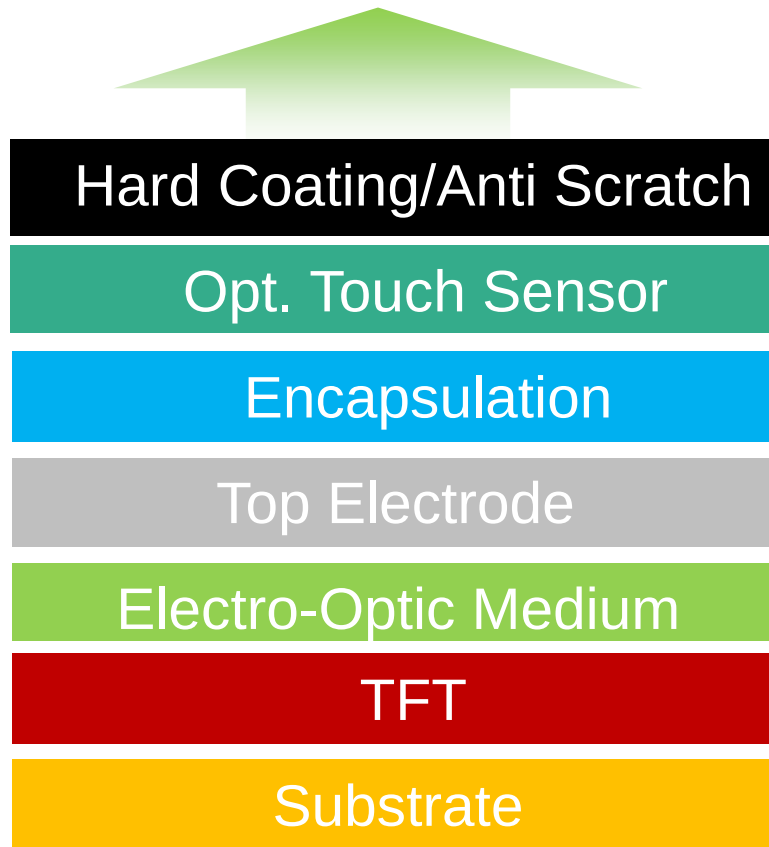
- Used as interconnects between rigid devices or in sensor structures, including stretch and impact sensors
- New figures-of-merit for inks such as stretchability, washability, lack of fatigue, etc

Applications include:

- Impact sensor by BeBop. Here 6 layers of various stretchable inks are printed
- Respiration sensors detecting changes in dimensions of the conductive track induced by breathing .This is by EMS
- Activity tracking functionality (HRM, EMG, etc) by Clothing+ and DuPont



6. Plastic and Flexible Displays



- Differentiate (thinner, lighter weight, new applications possible)
- Plastic substrate and curved displays used in smart watches and phones
- Over 100 million electrophoretic displays sold – inherently flexible display plane (but finished device may not be)
- Many opportunities including flexible anti scratch layers.
- LG announced \$900million investment in flexible OLED displays in 2015



7. Diverse Landscape of Wirelessly Connected Things

Bought as a productivity/ connectivity tool

Personal Interface Electronics

Phones, tablets, watches and computers

2020: 2.6 Billion

Mainly bought by consumers

Usually payback driven or implemented for security and safety

Internet Connected (IP address)

Sense, learn, adapt, communicate and co-operate and control over the internet with no human interaction.

2020: 1.9 Billion

Mainly bought by Governments today
Utility/Energy and Healthcare strongest growth areas

ID and Sensor systems (no IP address or proxy)

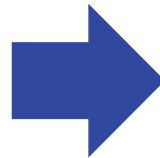
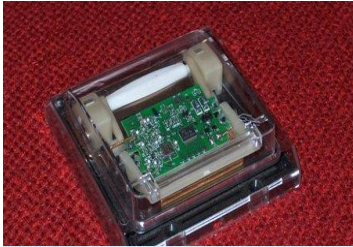
Proprietary or standardized RFID, Active RFID, Real Time Location Systems, Sensor Networks

2020: 25.4 Billion

Mainly bought by industry and Governments today

8. Energy Harvesting Goes Further

mW to W



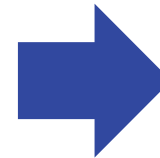
10 W to 100KW+



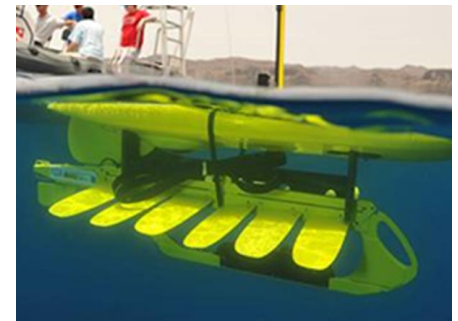
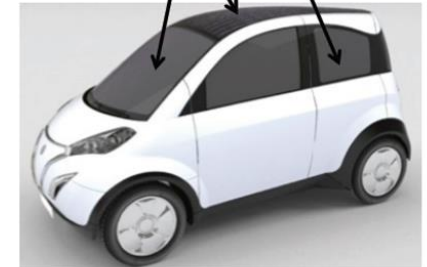
Regenerative
suspension



Waste heat
thermoelectrics



PV Surfaces



9. Sensor Proliferation

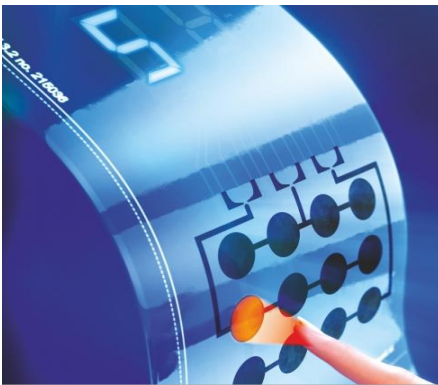
New Form Factors and Capabilities



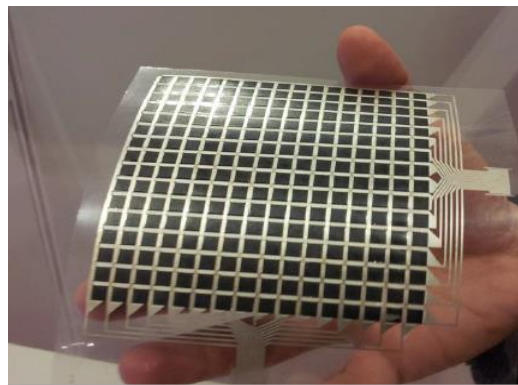
Source: ISORG



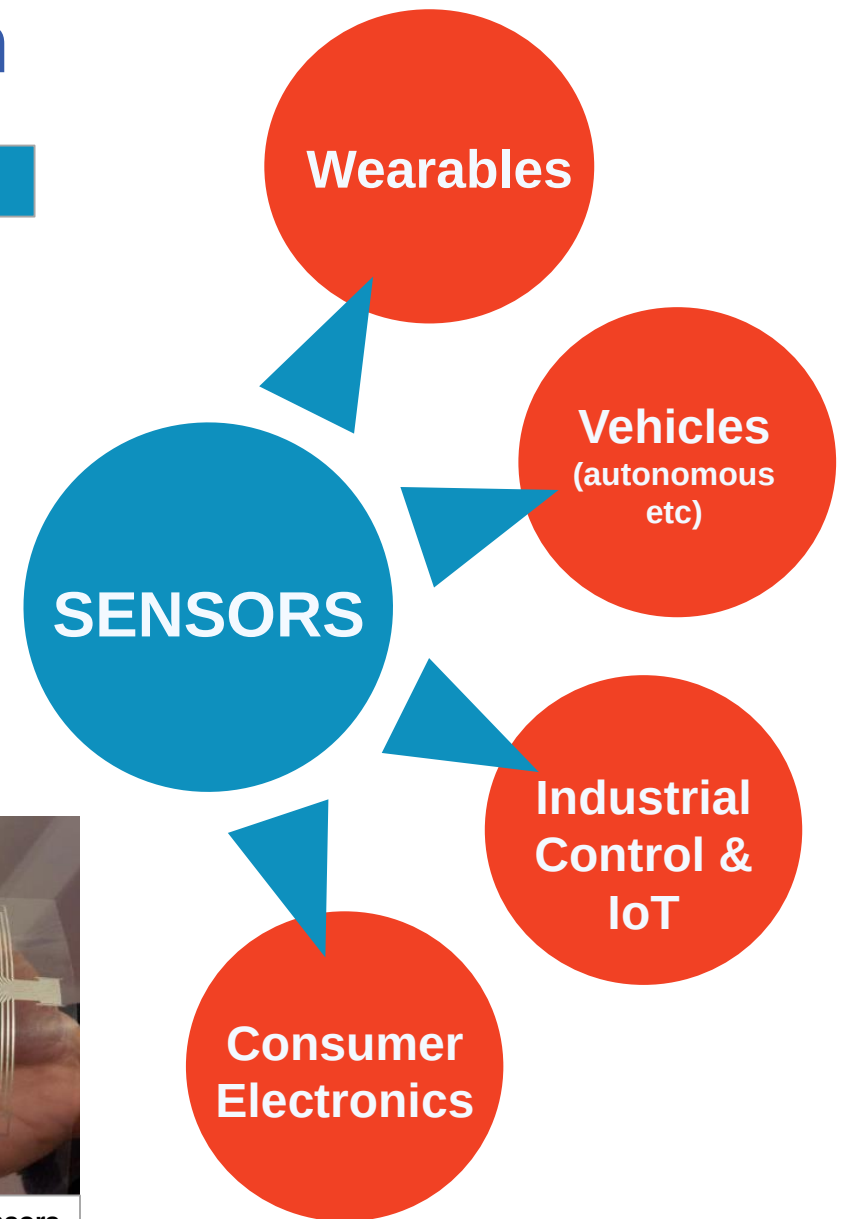
Source: IMEC and Holst Centre



Source: 3PLAST project



Printed thermistor array, PST Sensors



10. Advanced New Materials Come to Market

Conductive Inks

- Stretchable inks for e-textiles
- Mold or thermoform electronics
- Inks for 3D printing
- Mobile phone digitizers

Graphene

- Capacity increasing and price falling
- Energy storage will be one of the largest markets for graphene, but many others too

Quantum Dots

- Commercial in displays as part of the color filter for better color gamut
- Work on QDOT lamps to replace LED lamps

Perovskites

- High efficiency PV, potentially low cost device
- Working on stability, Pb-free versions & manufacturing scale-up
- Hybrid tandem devices to come first

Transparent Conductive Films

- Metal mesh, nanowire and other options in products today as an ITO replacement

Electroactive Polymers

- Haptics and transducers

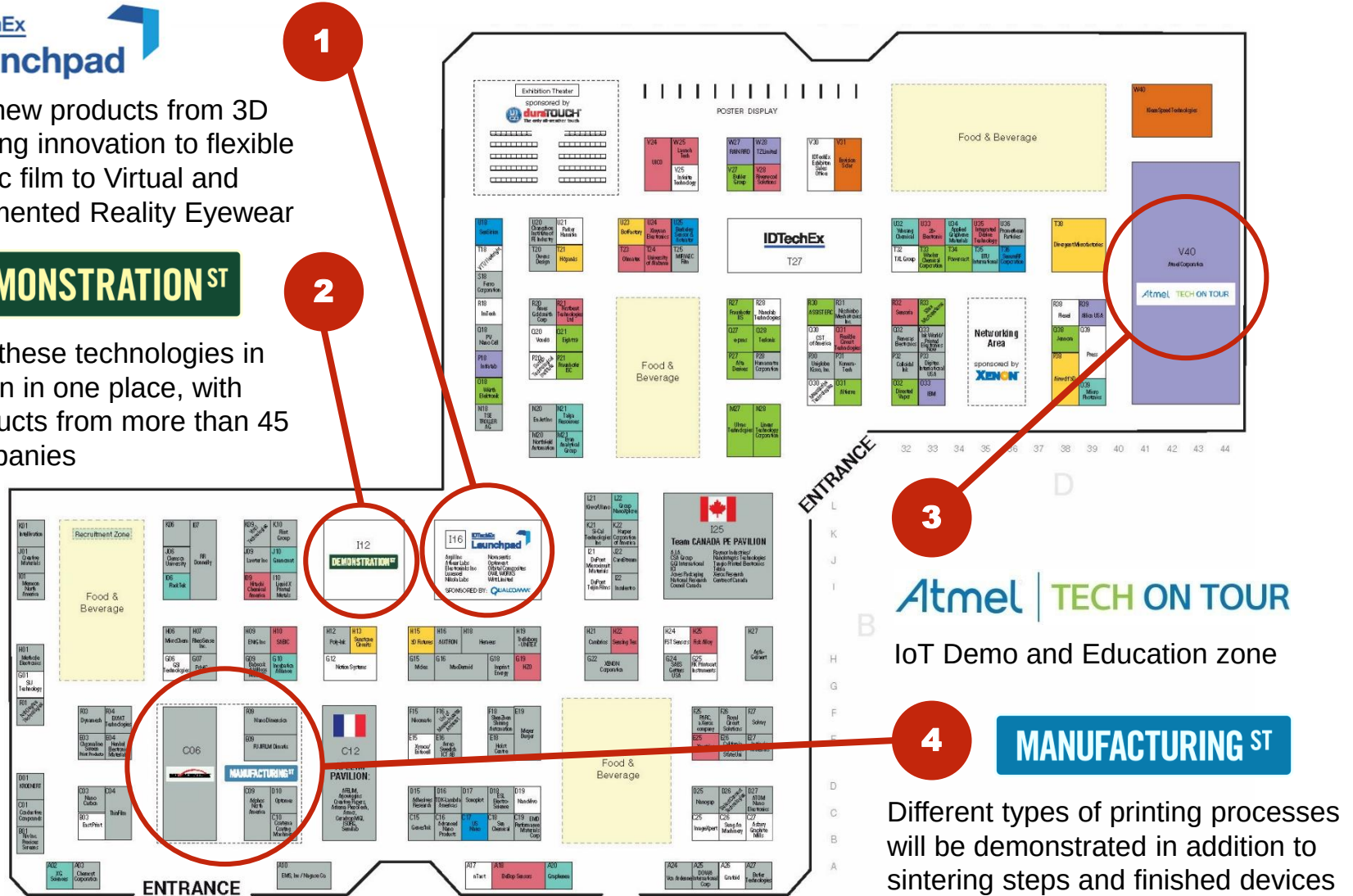
Products, Manufacturing and Much More to See

IDTechEx Launchpad

Ten new products from 3D printing innovation to flexible haptic film to Virtual and Augmented Reality Eyewear

DEMONSTRATION ST

See these technologies in action in one place, with products from more than 45 companies

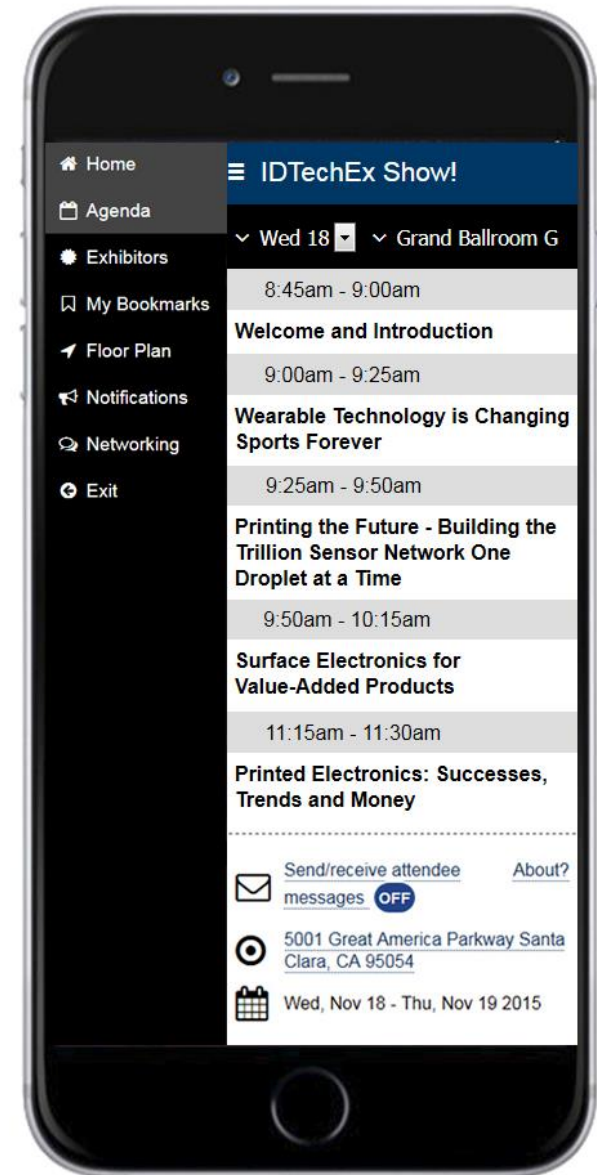


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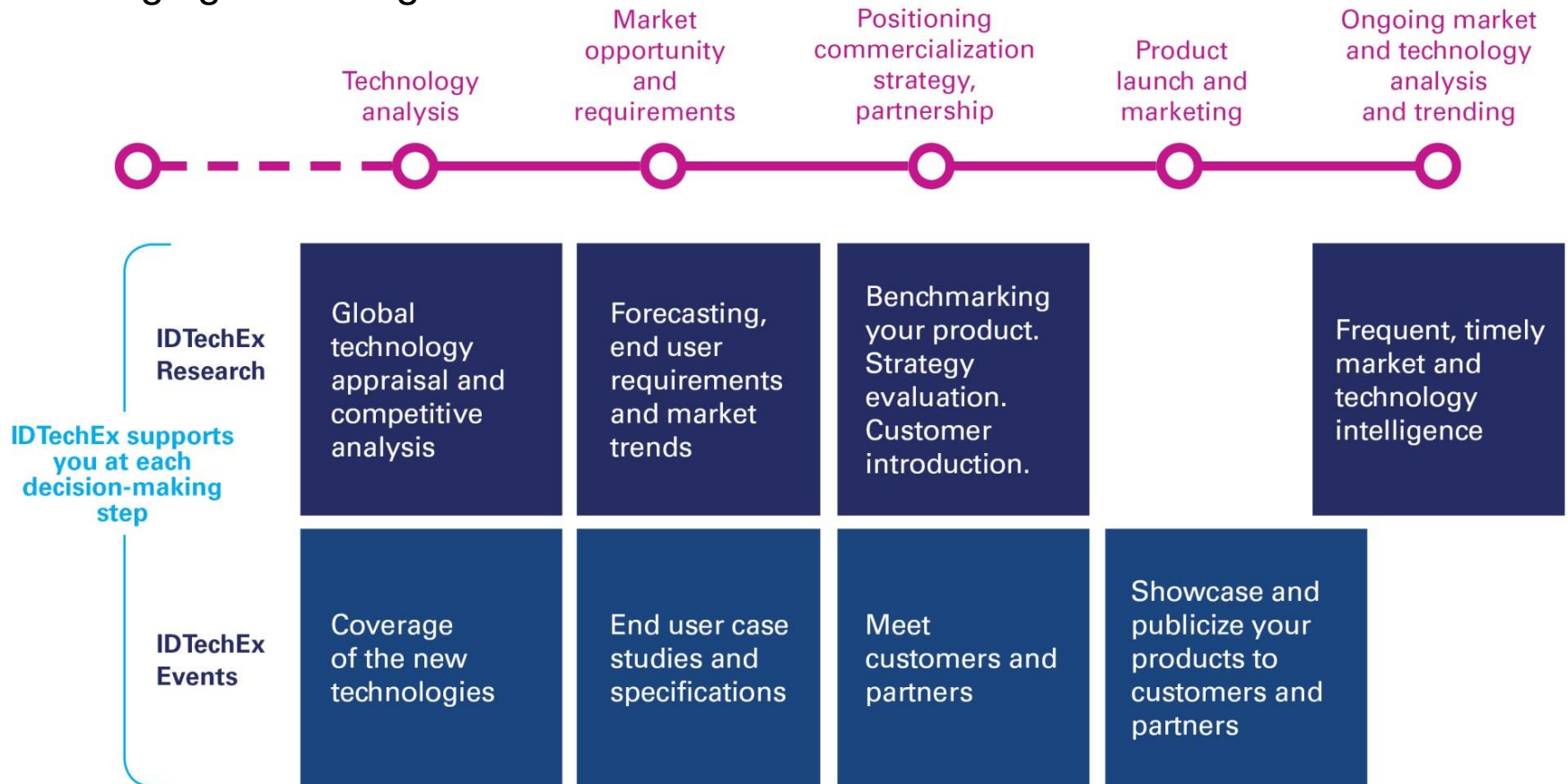


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Cornerstone and Keynote Agenda



**United States Olympic
Committee**



Hewlett Packard



Qualcomm

Tradeshow opening

